Linux VM Authentication Guide

This document explains the two authentication models supported by the `linuxvm` module:

- traditional Active Directory domain join
- Microsoft Entra ID SSH login

These models can coexist, but they are not the same thing and they do not use the same login path.

Authentication Modes

1. Active Directory Domain Join

This is the traditional enterprise model.

When `enable_domain_join = true`, the module:

- renders the domain join values into `custom_data`
- boots the VM and runs init.sh
- uses managed identity plus Key Vault to retrieve the domain join secret
- installs `realmd`, `sssd`, `adcli`, and related packages
- joins the VM to the AD domain
- configures guest SSH and sudo access for the requested groups

This path is guest-OS native. The VM becomes a domain-joined Linux host.

When `enable_domain_join = false`, the module still runs the normal `init.sh` bootstrap, but the script now skips the entire AD join path cleanly without attempting Azure login, Key Vault secret retrieval, or `realm join`.

2. Microsoft Entra ID SSH Login

This is the Azure-native SSH model.

When `enable_entra_ssh_login = true`, the module:

- installs the `AADSSHLoginForLinux` VM extension
- adds Azure VM login RBAC for the provided groups
  - `app_admin_group` -> `Virtual Machine Administrator Login`
  - `app_user_group` -> `Virtual Machine User Login`
- keeps the existing VM resource RBAC
- `app_admin_group` -> `Contributor` on each VM resource, `Contributor` on each module-created NIC, plus `Reader` on the VM resource group
- `app_user_group` -> `Reader` on each VM resource, plus `Reader` on the VM resource group

When Bastion is configured:

- `app_admin_group` -> `Reader` on the Bastion resource group and `Network Contributor` on the Bastion host
- `app_user_group` -> `Reader` on the Bastion resource group and `Network Contributor` on the Bastion host

For the VM resource group and Bastion resource group `Reader` assignments, the module checks for existing assignments at that exact resource-group scope so consumers can import matching ones from the root module or with `terraform import` before apply. Reader assignments on child resources under the resource group do not satisfy the resource-group Reader path.

This path is Azure identity based. It does not require the VM to be domain joined.

Time Sequence

The authentication pieces are applied in this order:

1. Terraform creates the Linux VM.
2. Terraform passes the rendered `custom_data` payload to the VM.
3. On first boot, cloud-init runs init.sh.
4. If `enable_domain_join = true`, the AD join steps run inside the guest.
5. Terraform installs the optional `AADSSHLoginForLinux` extension when `enable_entra_ssh_login = true`.
6. Terraform applies the Azure RBAC role assignments for VM resource access and, when enabled, Entra VM login access.

This means:

- AD authentication is established through guest configuration in `custom_data`
- Entra SSH authentication is established through the VM extension plus Azure RBAC

Use Cases

Use AD Domain Join When

- the VM must behave like a traditional domain-joined server
- Linux identity resolution should come from AD through `sssd`
- guest sudo access should be driven by domain groups
- existing operational tooling expects classic AD integration

Use Entra SSH When

- the VM is not domain joined
- access should be controlled through Azure RBAC
- administrators connect from Azure CLI, Bastion, or other Entra-aware tooling
- the target operating model is cloud-native access rather than classic domain trust

Use Both When

- you are in a migration period
- you need both classic AD behavior and Entra-based SSH access
- different teams use different access methods temporarily

Using both is valid, but it increases support and troubleshooting complexity.

AD Login Details

For the domain-join path, the important module input is:

- `enable_domain_join = true`

Important prerequisites:

- system-assigned managed identity is enabled on the VM
- the VM can reach Azure endpoints and the domain infrastructure
- the shared Key Vault contains the domain join password
- DNS and network routing support the domain join path

Expected user experience:

- users connect with normal SSH methods supported by the guest
- authentication is resolved through the joined AD domain
- `app_admin_group` members can receive guest sudo access through the script configuration

Entra SSH Details

For the Entra path, the important module input is:

- `enable_entra_ssh_login = true`

Important prerequisites:

- the `AADSSHLoginForLinux` extension is installed successfully
- the user or group has the right Azure RBAC assignment on the VM
- the client uses an Entra-aware SSH flow
- the VM can reach required Azure and Entra endpoints over HTTPS

RBAC Mapping

When Entra SSH is enabled, the module adds:

- `app_admin_group` -> `Virtual Machine Administrator Login`
- `app_user_group` -> `Virtual Machine User Login`

These are guest sign-in roles.

They are different from:

- `Contributor`
- `Reader`

Those two are Azure resource management roles, not VM guest login roles.

Group Input Guidance

For `app_admin_group` and `app_user_group`:

- group object IDs are preferred
- group display names are also supported
- display-name resolution uses Microsoft Graph through the `azuread_group` data source
- the Terraform identity therefore needs sufficient Microsoft Graph read permission when display names are used
- user object IDs are not a supported replacement for group IDs in these inputs

Working Entra Methods

These methods are expected to work when the extension, RBAC, and networking are correct:

- `az ssh vm`
- `az ssh config` followed by `ssh`
- Azure Bastion with Microsoft Entra ID
- Azure portal SSH when the authentication type is switched to `Microsoft Entra ID`

Methods That Are Not Expected To Work

These are common paths that should not be treated as the supported Entra SSH method:

- plain `ssh <aad-user>@<host>` with no Azure CLI-assisted Entra flow
- Azure portal SSH with `VM password` when you expect Entra credentials to work
- using `Contributor` or `Reader` alone and expecting guest login access
- assuming a synced on-prem AD password is the same as Linux local VM password authentication

Why `VM password` Does Not Equal Entra Login

In the Azure portal, `Microsoft Entra ID` and `VM password` are different authentication paths.

`Microsoft Entra ID`:

- uses the `AADSSHLoginForLinux` extension
- uses Azure and Entra authentication
- depends on `Virtual Machine Administrator Login` or `Virtual Machine User Login`

`VM password`:

- uses the VM's local Linux account
- expects the local `admin_username` and `admin_password`

- does not use your Entra identity

In this module, the Entra extension settings also disable SSH password authentication for the Entra path. So an Entra user succeeding with `Microsoft Entra ID` and failing with `VM password` is expected behavior.

Administrator vs User Login Roles

`Virtual Machine User Login`

This role is for regular Entra-based sign-in to the VM.

It is intended for:

- non-admin SSH access
- basic shell access
- operations that do not require elevation

`Virtual Machine Administrator Login`

This role is for elevated Entra-based sign-in to the VM.

It is intended for:

- administrator SSH access
- users who need elevated privileges on the VM

For Linux, this is the Azure role associated with admin-level sign-in and expected `sudo` capability after logon.

Troubleshooting Checklist

If Entra login still fails:

1. Confirm `enable_entra_ssh_login = true`.
2. Confirm the VM extension is installed and healthy.
3. Confirm the user or group has `Virtual Machine Administrator Login` or `Virtual Machine User Login`.
4. Wait for RBAC propagation.
5. If the role is eligible through PIM, activate it.
6. Use `az ssh vm` or `az ssh config` instead of plain `ssh`.
7. Confirm port `22` access through the intended network path.
8. Confirm outbound HTTPS from the VM to Azure and Entra endpoints.
9. Check Conditional Access or MFA requirements.

If AD login fails:

1. Confirm `enable_domain_join = true`.
2. Confirm the VM has managed identity enabled.
3. Confirm Key Vault access is working.

4. Confirm DNS and domain controller reachability.
5. Review guest logs from the bootstrap phase.

Related Docs

- README.md
- INIT.md
- LOCALIZATION.md